

Weaving Te Ao Māori into Engineering Practice

Kahu Jones

4th Year Computer Engineering Student, University of Canterbury

ENEL301 Tutor

Co-presented with Andre Konia Meridian Energy



Why This Matters for Engineering in Aotearoa

1

Te Tiriti o Waitangi

A partnership framework, not a historical document - it shapes how engineers work with communities today.

2

Graduate attributes

Engineering NZ and Washington Accord expectations increasingly include cultural competence, not just technical skill.

3

Practical, not abstract

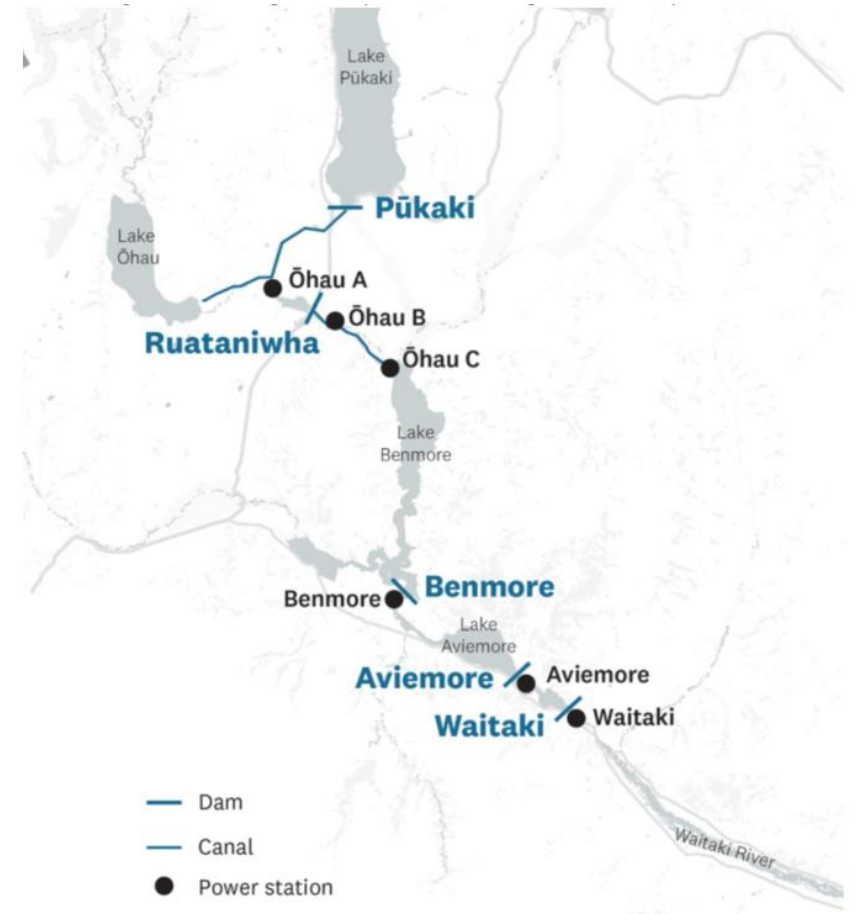
It shows up as who you consult, whose knowledge counts as data, and whose success criteria you design for.

Any engineer working on land, water, or infrastructure in NZ will meet mana whenua relationships at some point.



Infrastructure Built on Whenua and Awa

- Meridian's hydro stations don't just sit near rivers - the awa are the resource being used.
- Iwi and rūnanga hold enduring relationships with these specific waterways that predate the power stations by centuries.
- Cultural engagement isn't a side initiative - it sits inside engineering planning, environmental management, and now, digital innovation.



The Real Constraint: Access

- Hydro sites are hazardous, remote, and access-restricted: confined spaces, high-energy systems, water hazards.
- Kaumātua and whānau with generational ties to a specific awa may never set foot on the site their tūpuna knew.
- **An engineer who only sees “hazard = restrict access” has solved the safety problem - and created a cultural one.**





Meridian Hydro Vision

Goal –

Transform existing site data into a user-friendly VR platform for accessible and immersive exploration.

- Meridian already had high-resolution 3D scans of hydro assets, used by engineers for inspection and planning.
- The pivot: same underlying data, completely different audience and purpose - not inspection, but access and connection.
- Guided by health & safety, communications, and cultural/environmental information.



Whānau and Kaumātua as the Design Target

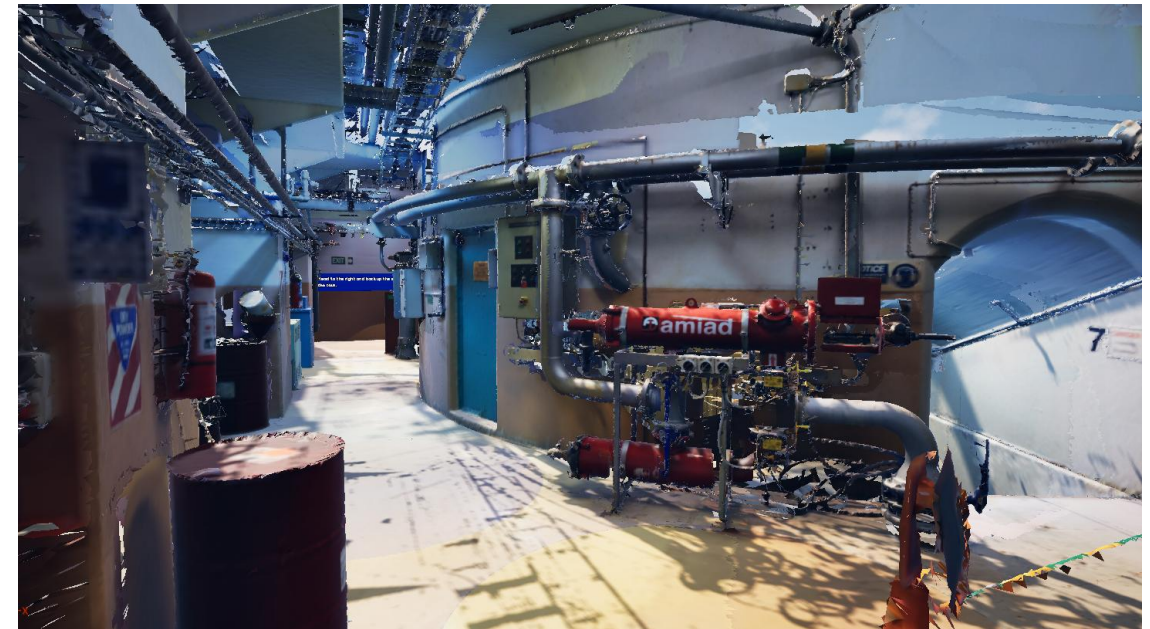
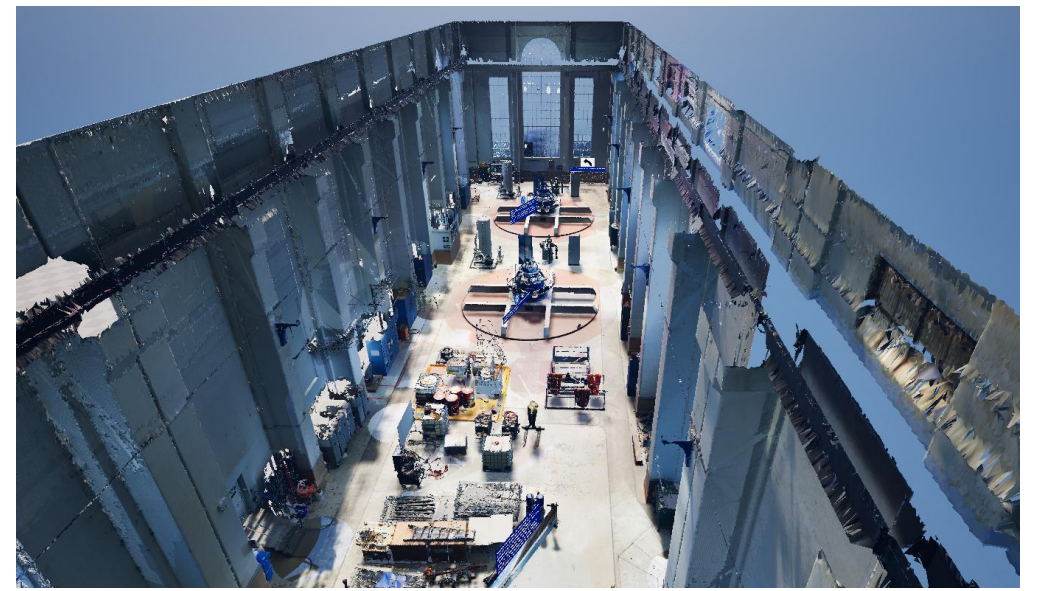
“Focusing on whānau and kaumātua who would never get to see the hydro sites on their awa.”

- Interaction mechanisms simplified for first-time VR users, not power users.
- Visual clarity prioritized over photorealism
- Comfort and understanding mattering more than technical showmanship.



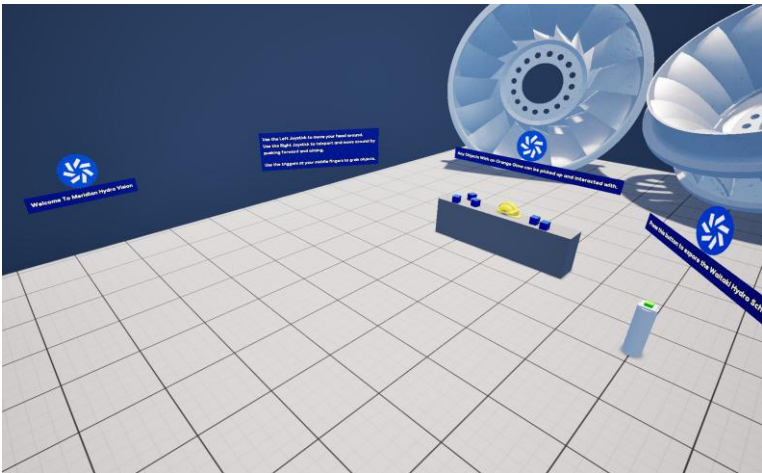
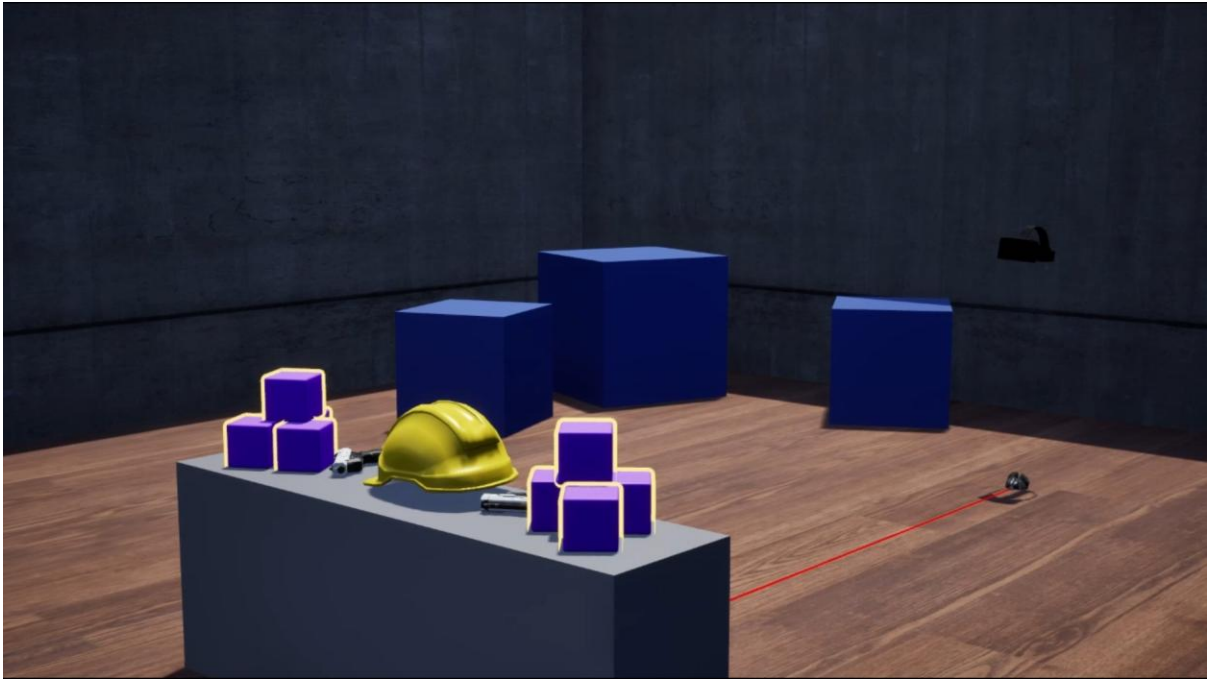
THE OUTCOME

Meridian Hydro Vision



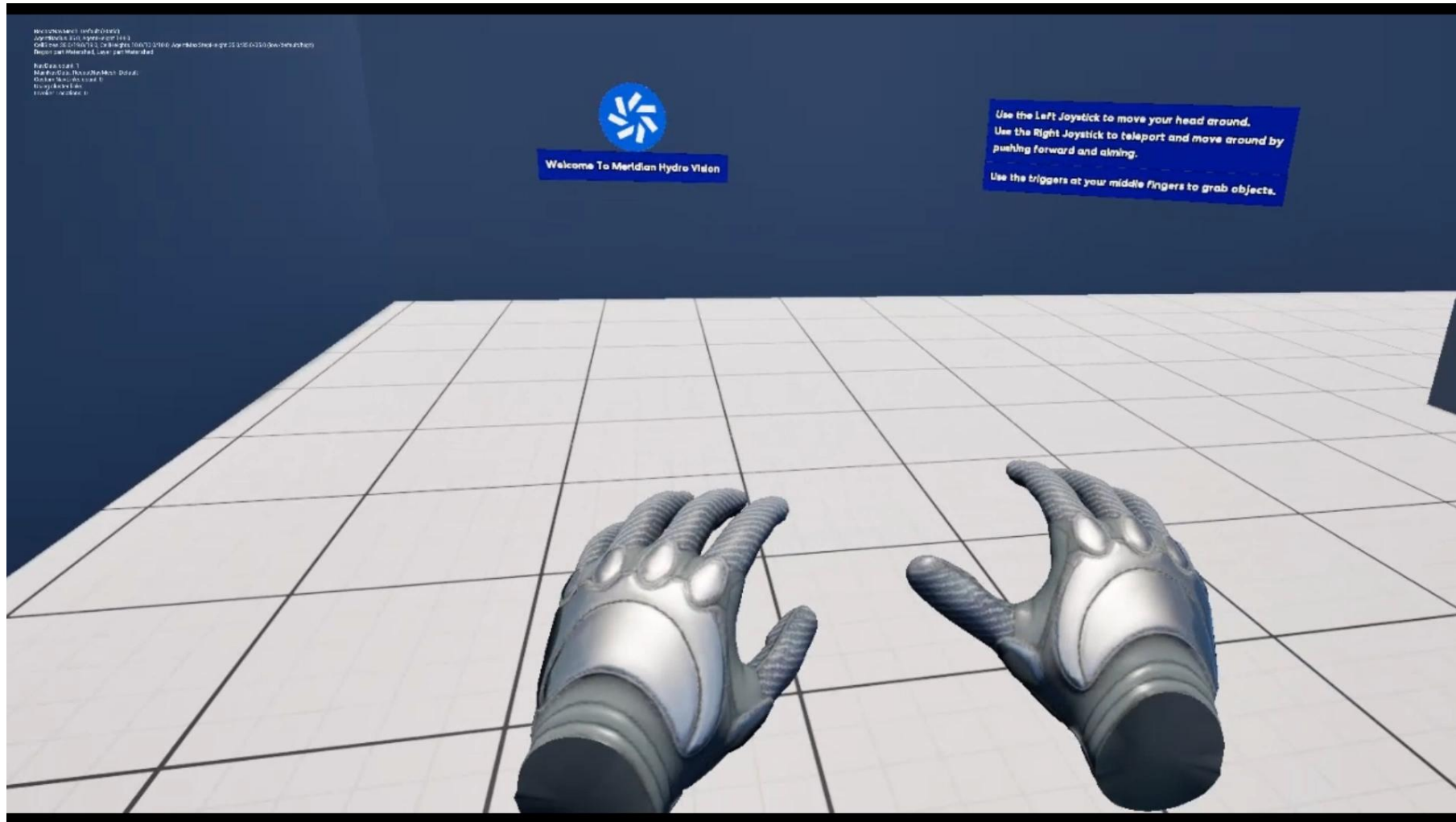
THE OUTCOME

Meridian Hydro Vision



THE OUTCOME

Meridian Hydro Vision – Prototype



Tuna Trap & Transfer Program

- Tuna (eels) are both an ecological indicator species and a taonga species - mahinga kai, traditional food gathering.
- Fish passage and river management decisions intersect directly with these cultural values.
- Site visits framed by manaakitanga: hospitality, care, and reciprocity - not a formality before “real” engineering starts.
- Relationship-building is the mechanism through which engineers learn what actually matters to those affected by their work.



“Biculturalism isn’t a section of the report – it’s a design input.”

- Whatever discipline you end up in, you will eventually touch whenua, water, or a community with a prior relationship to that place.
- Ask who else has a stake before you scope the solution - not after.
- Here, that question changed the audience, the interaction design, and the success criteria for the whole project.



He aha te mea nui o te ao

What is the most important thing in the
world?

He tangata, he tangata, he tangata

It is the people, it is the people , it is the
people
- Māori Proverb

Discussion / Questions - He Pātai?

Think of an engineering project you've worked on — who was affected by it that wasn't in the room when it was scoped?

How would you find out you'd missed a stakeholder like this, before the project was finished rather than after?

